MalawiDrop

1. Schematic Design of device

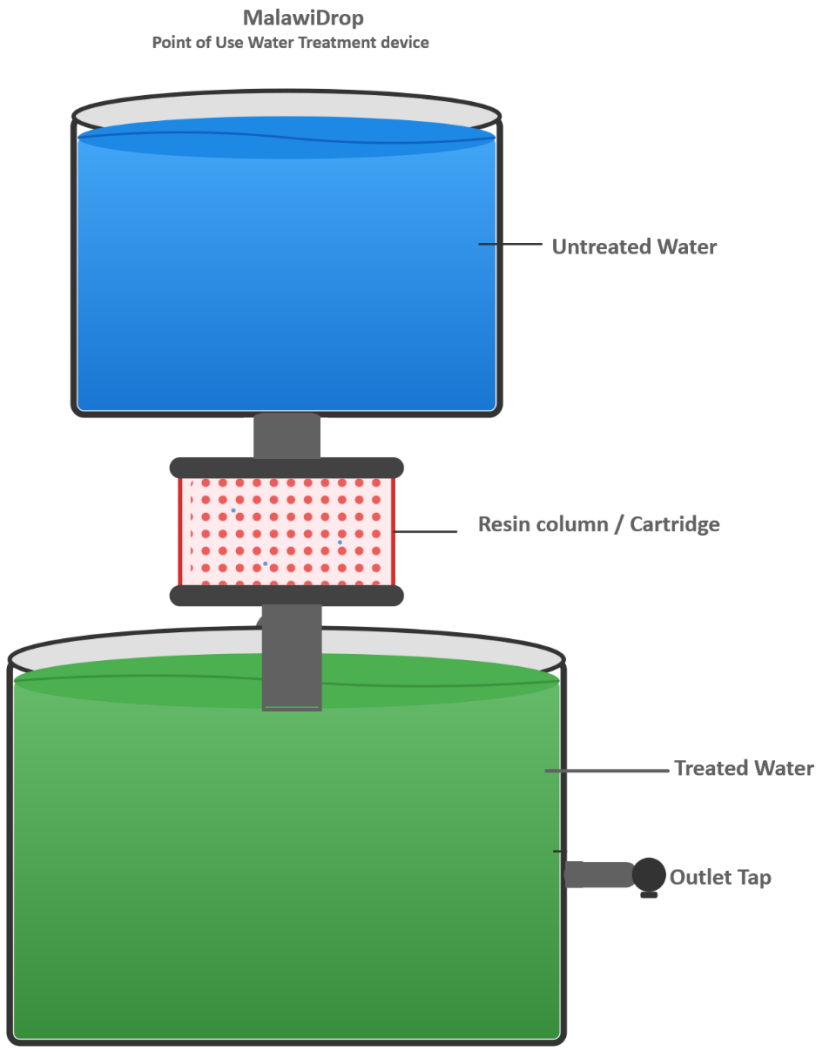

Figure 1 Two bucket resin based disinfection unit design

2. Lab-Scale Prototype

Figure 2 Two bucket system testing efficacy of disinfection of resin column at Lab scale

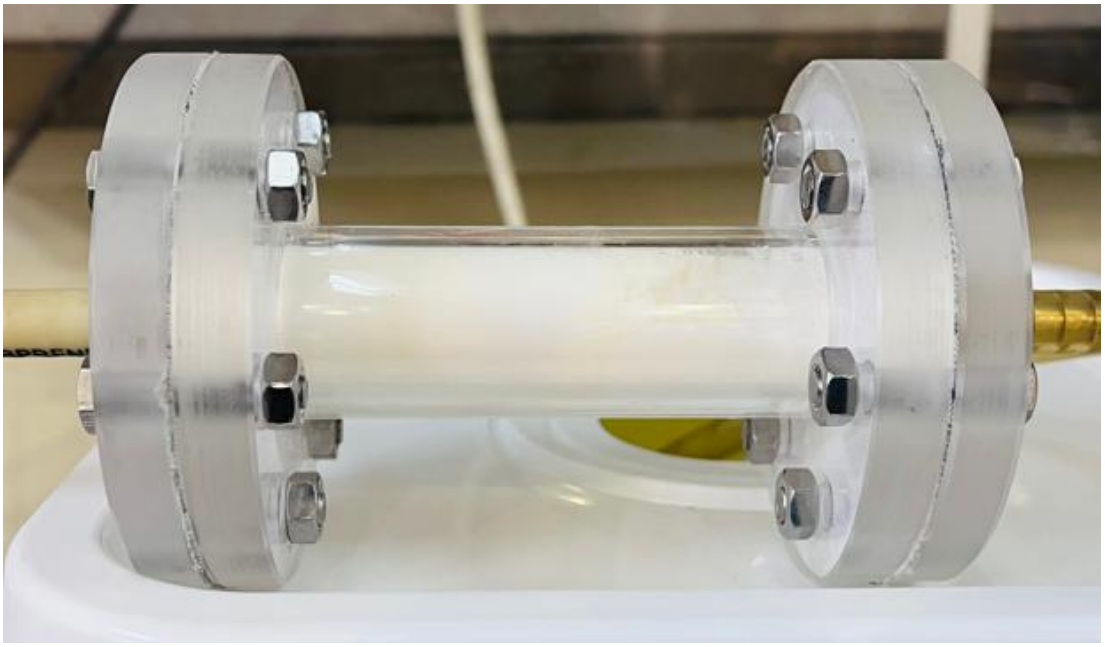

Figure 3 Lab Scale Resin Column/Cartridge Prototype

3. Prototype Zero

Figure 4 Low-Cost Prototype Zero

4. Pilot

5. Performance Results

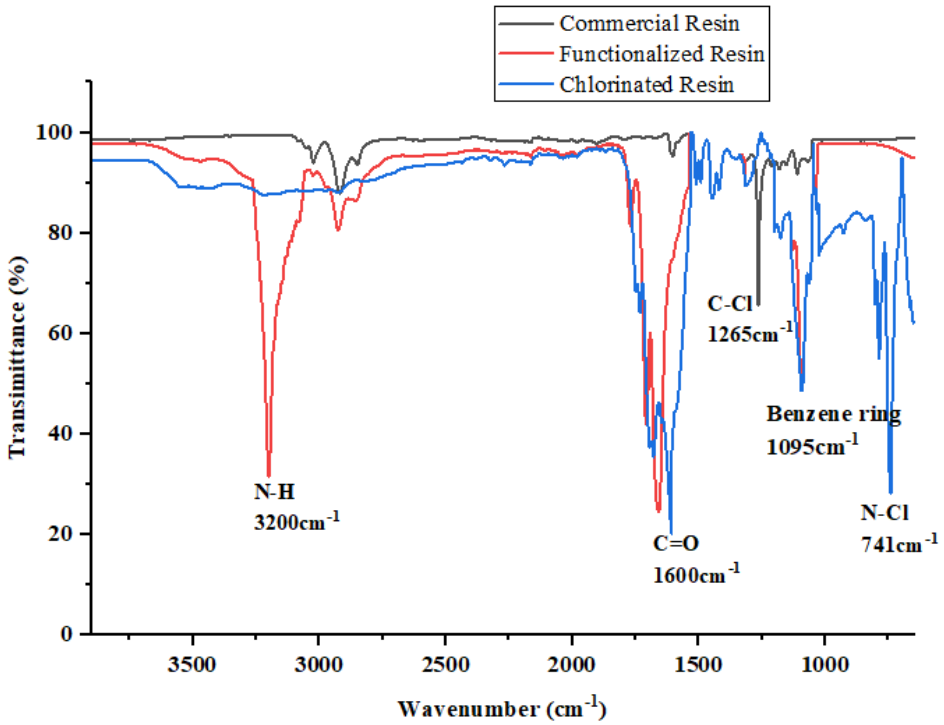

Figure 5 represents the FTIR spectra for the resin before the grafting reaction with DMH-K, after the grafting reaction and the spectra of the chlorinated resin. The presence of the N-Cl band indicates the effective chlorination of the resin, thus the synthesis

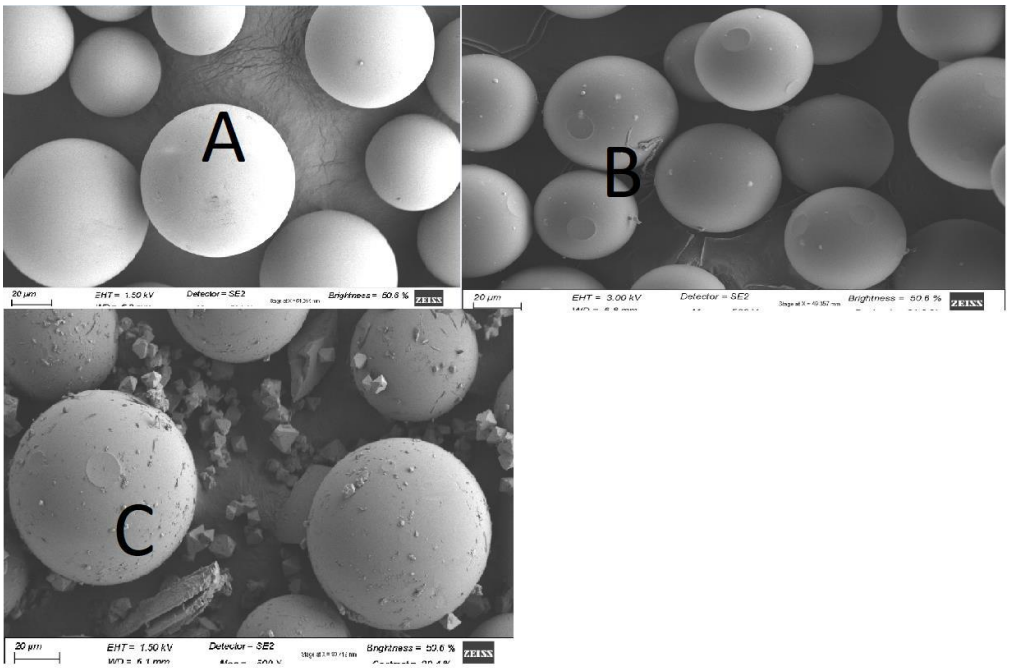

Figure 6 SEM Images of (A) unreacted resin, (B) modified resin in DMSO, (C) modified resin in DMF

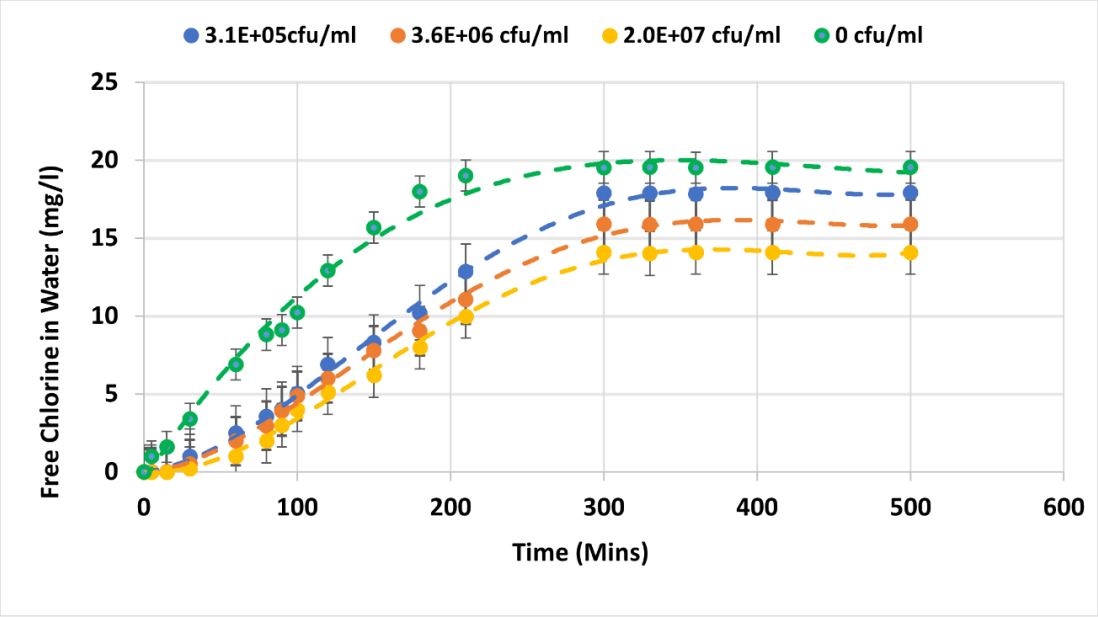

Figure 7 Free chlorine measured in water with a varying bacteria concentration and a constant resin mass ; cfu/ml representing amount of bacteria in water.

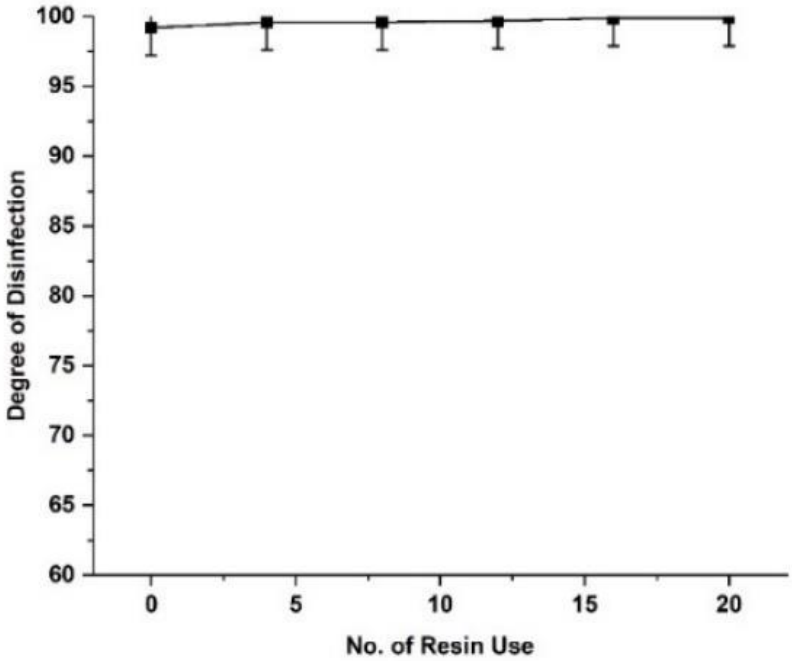

Figure 8 Disinfection efficacy of resin after 20 cycles demonstrating stability of engineered resin